



Appium Installation for Windows

1. Make sure that you have JDK, version 8 or a later version needs to be installed <https://www.oracle.com/java/technologies/downloads/>

Likely you already have it since we have been using it for Java coding. To make sure that you have a valid JDK version, in command prompt type the following instruction and confirm the output.

```
java --version
java 21.0.2 2024-01-16 LTS
Java(TM) SE Runtime Environment (build 21.0.2+13-LTS-58)
Java HotSpot(TM) 64-Bit Server VM (build 21.0.2+13-LTS-58, mixed
mode, sharing)
```

The version could be different, but it should be 8 and later.

2. You should make sure that JAVA_HOME variable is properly set up. If you've already set it up, proceed to step 3.

To set up JAVA_HOME, follow these steps:

- a. Locate your Java installation directory. If you have not change the path during installation, it will be something like:

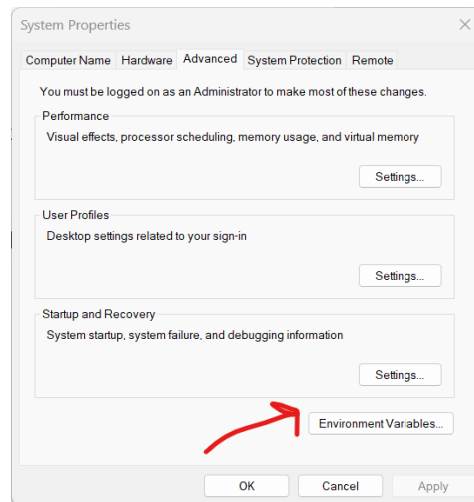
```
C:\Program Files\Eclipse Adoptium\jdk-11.0.17.8-hotspot\ or
C:\Program Files\Java\jdk-17.0.4.1\
```

You can also type `where java` at the command prompt, and you should see the full path of `java.exe` file.

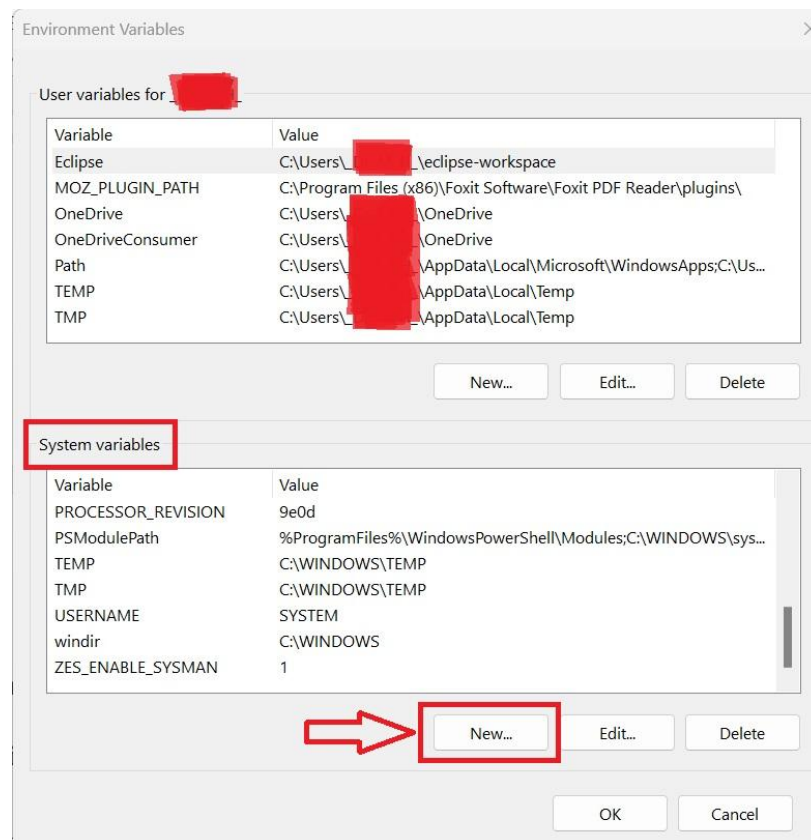
```
C:\Program Files\Java\jdk-21\bin\java.exe
```

NOTE: The above paths are examples! Make sure to properly locate JDK on your system.

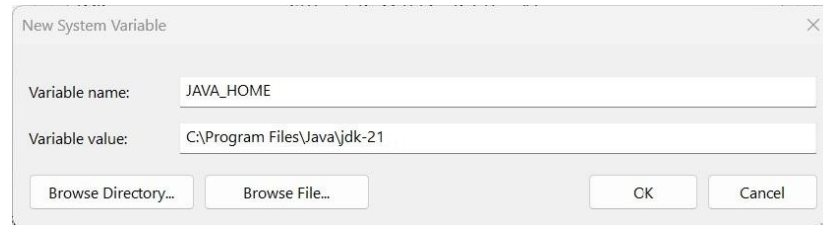
- b. Assuming that you have Windows 10 or 11, search for **Environment Variables** then select **Edit the system environment variables**
- c. Click the **Environment Variables** button.



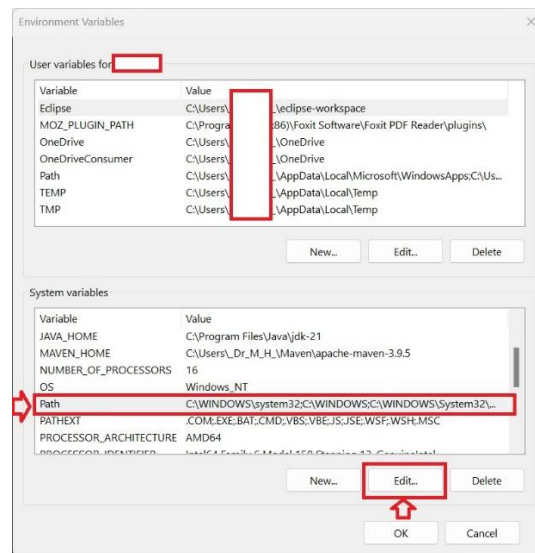
d. Under **System Variables**, click **New**.



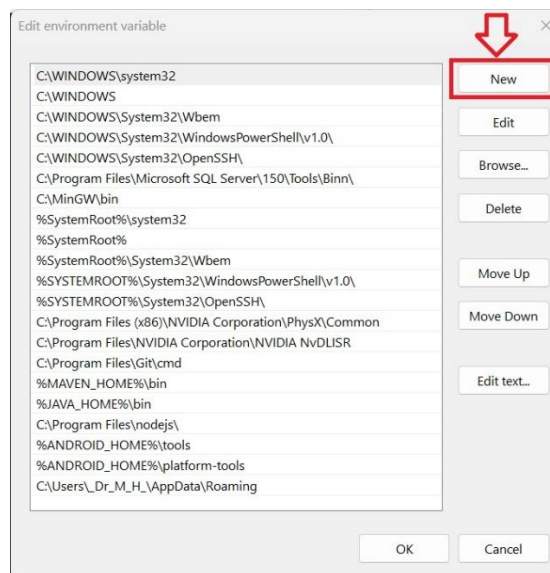
e. In the **Variable Name** field, enter `JAVA_HOME`, and in the **Variable Value** field, enter your JDK installation path.



- f. Click OK, and apply the changes.
- g. Under System Variables Sections, scroll down and find Path



- h. Select Path and then click Edit



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- i. In the new pop-up window click New
- j. Add the following entry and click OK

```
%JAVA_HOME%\bin
```

The following section can be skipped if JAVA_HOME is set properly via Windows.

Alternatively, you can directly set JAVA_HOME variable from the command line. To do so, follow these steps:

1. Make sure you open the Command Prompt as administrator.
2. Set the value of the environment variable to your JDK installation path using the following instruction:

```
setx /m JAVA_HOME "C:\Program Files\Java\jdk-21"
```

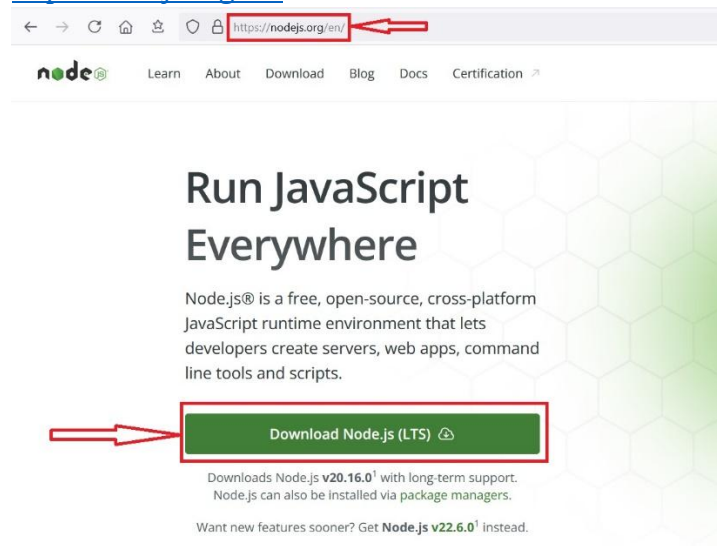
NOTE: Always restart the Command Prompt to reload the environment variables then use the following command to check if the JAVA_HOME it has been added correctly:

```
echo %JAVA_HOME%
```

You should see the path to your own JDK installation.

```
C:\Program Files\Java\jdk-21
```

3. As JavaScript is needed for running test cases in Appium, you should install the latest NodeJS from here <https://nodejs.org/en/>



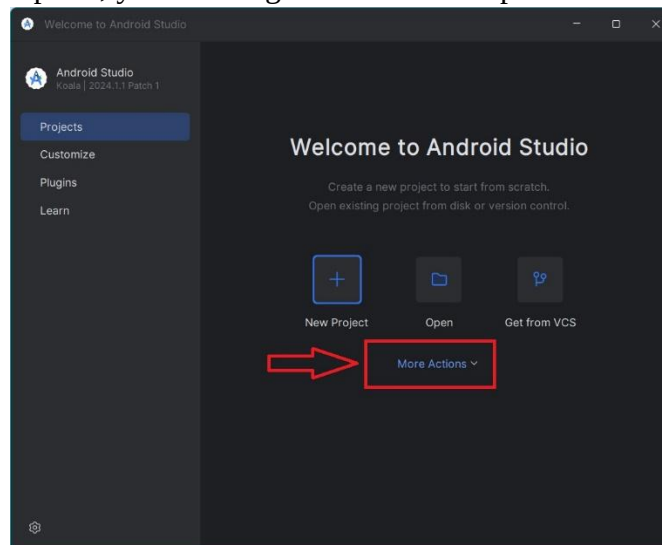
4. Once installed, run the following instructions to verify installation:

```
node -v  
v20.15.1  
  
npm -v  
10.7.0
```

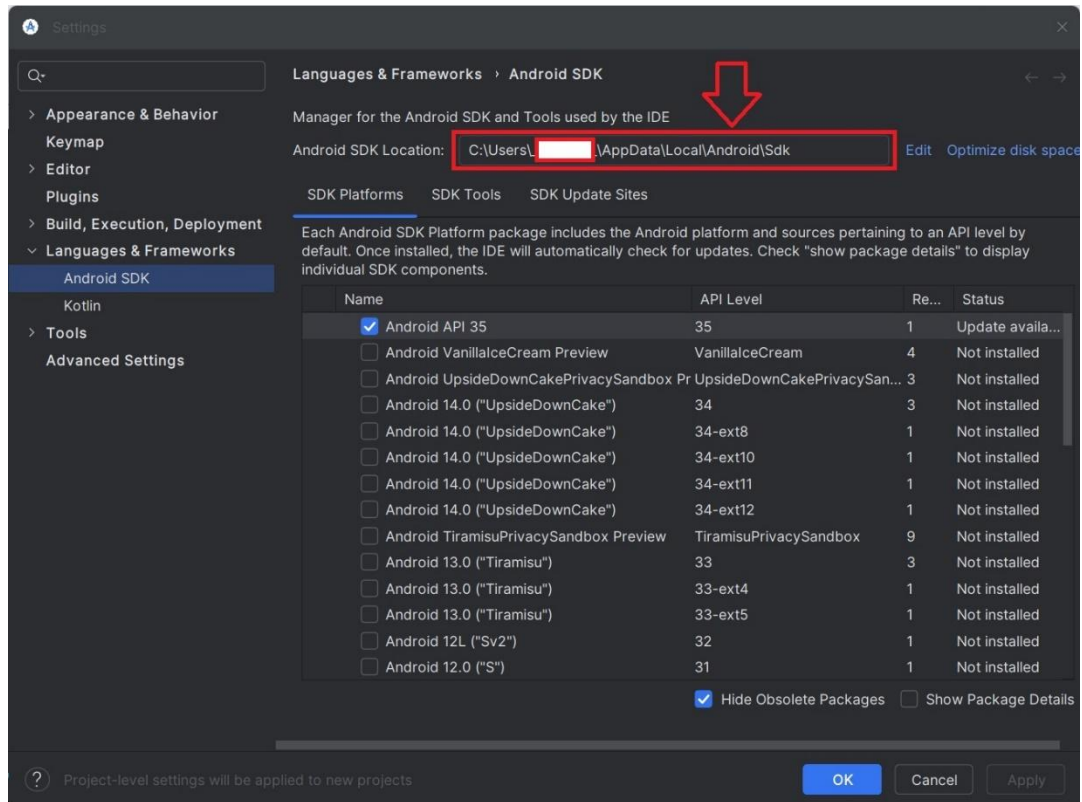
NOTE: The above versions may differ depending on the installed package.

5. Install Android Studio <https://developer.android.com/studio>

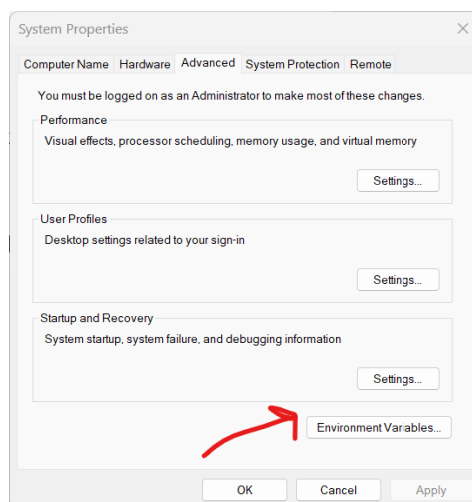
- Click Next, Next... → Install → Next,..., Finish
- Select Do not import settings and click OK
- Click Do not send → Next
- Select Standard for Install Type → Next → Next
- For the license, select the options one by one and select Accept → Click Finish
Wait for the download to complete
- Click Finish. At this point, you should get this Window opened:



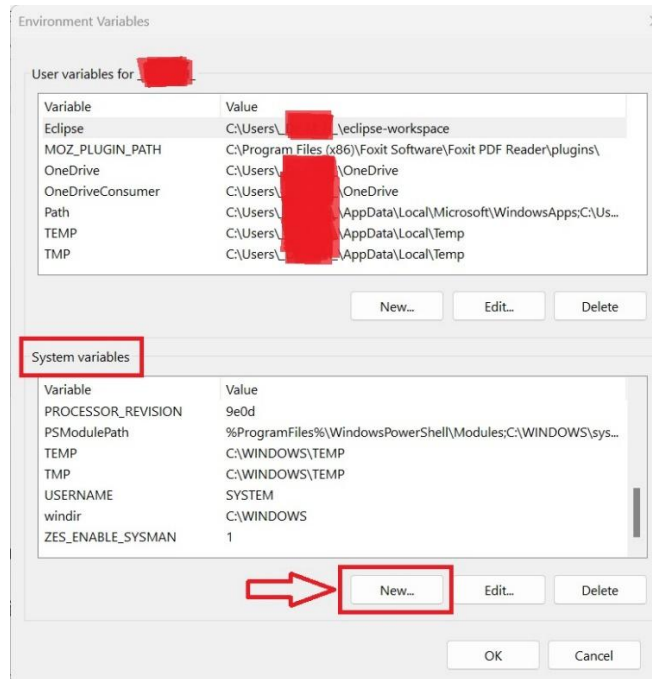
- Expand More Actions and select SDK Manager
- Copy the SDK installation location



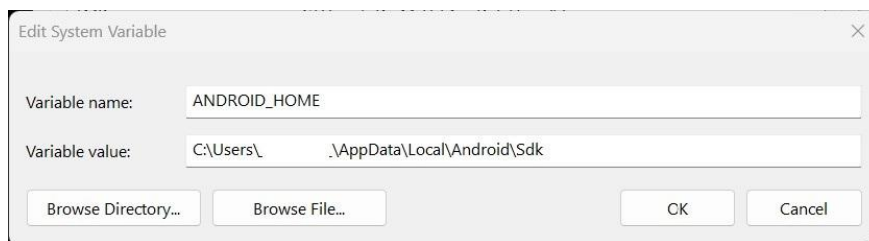
- Let us set the `ANDROID_HOME` variable
- From the start, type Environment Variables and open Edit the system environment variables, and click the **Environment Variables** button.



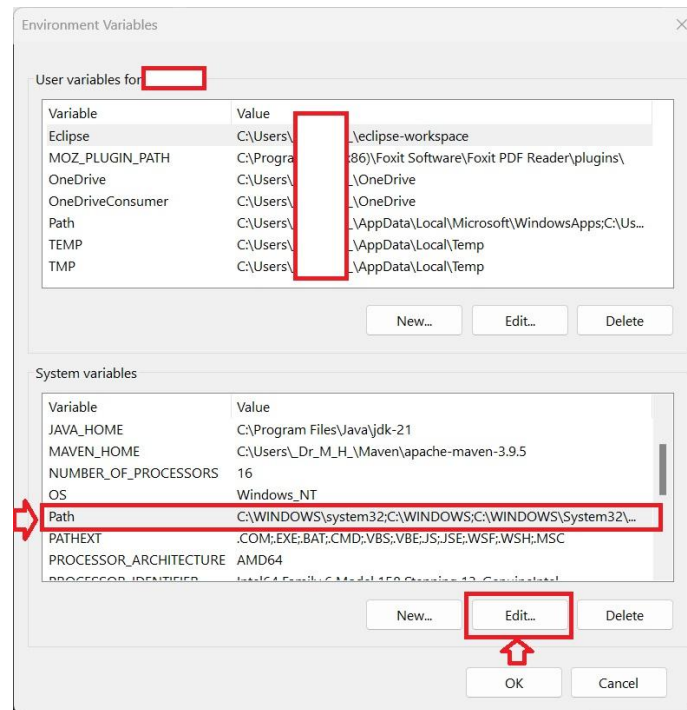
- Under **System Variables**, click **New**.



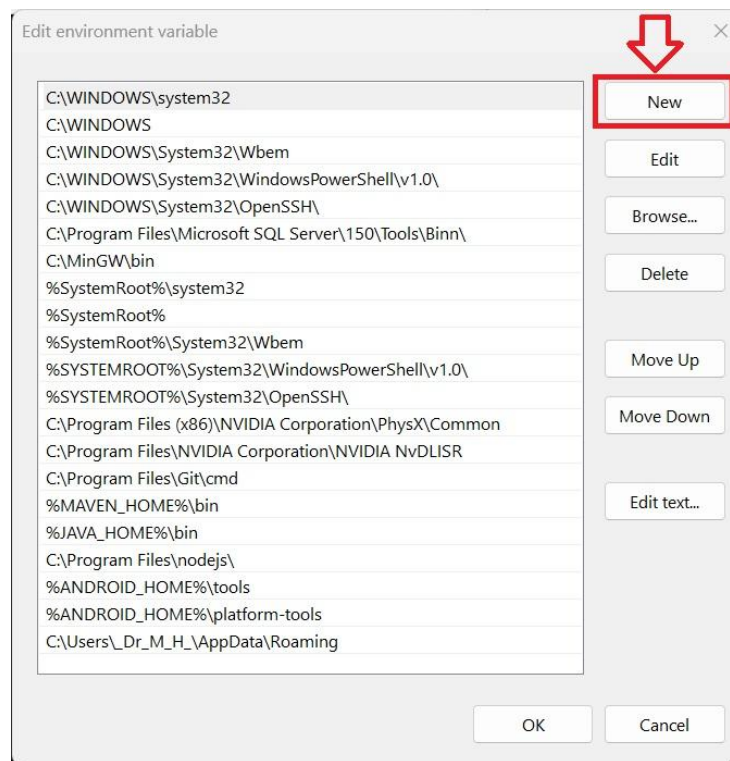
- k. In the **Variable Name** field, enter `ANDROID_HOME`, and in the **Variable Value** field, enter your SDK installation path.



- l. Click OK, and apply the changes.
- m. Under System Variables Sections, scroll down and find Path



n. Select Path and then click Edit



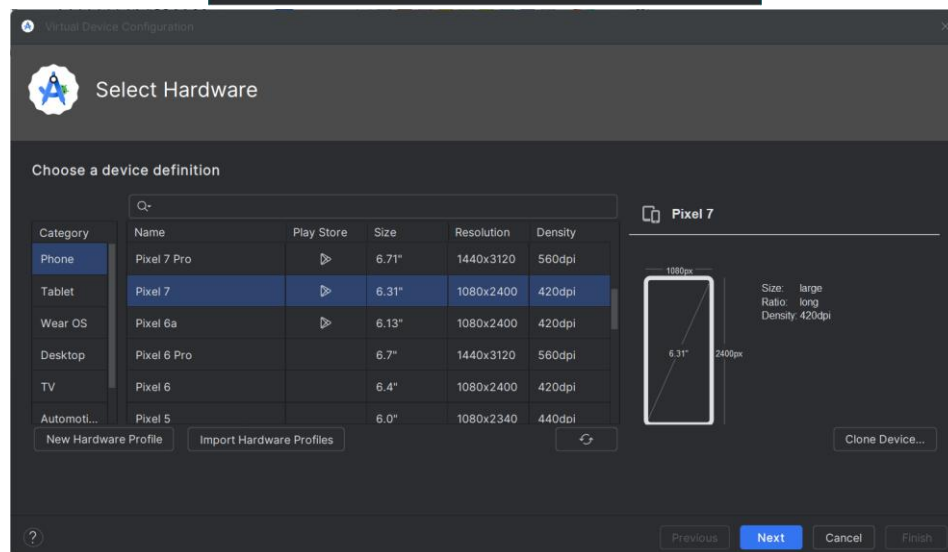
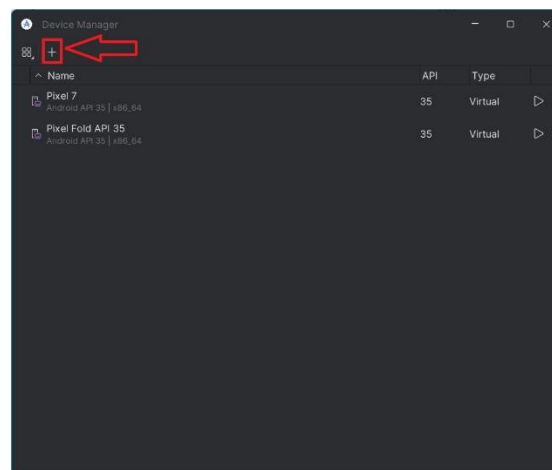
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- o. In the new pop-up window click New
- p. Add the following entries and click OK

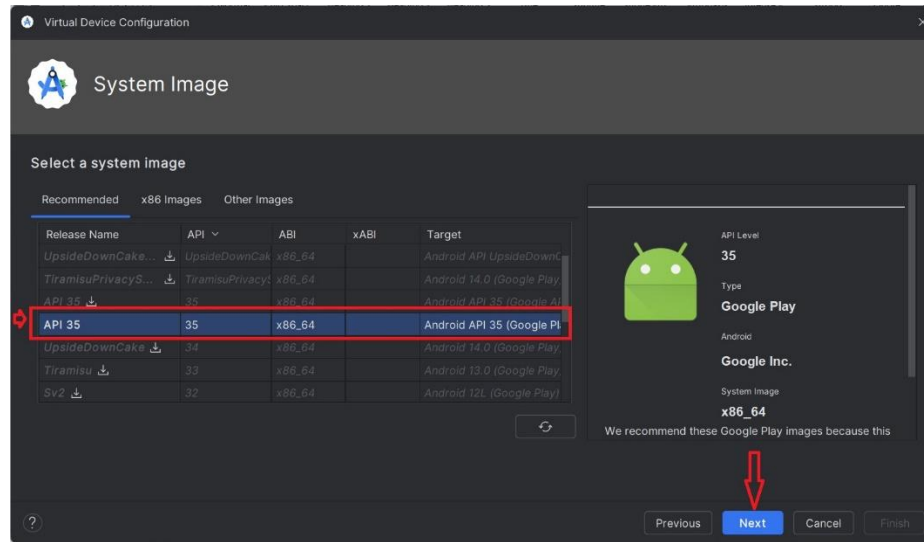
%ANDROID_HOME%\tools

%ANDROID_HOME%\platform-tools

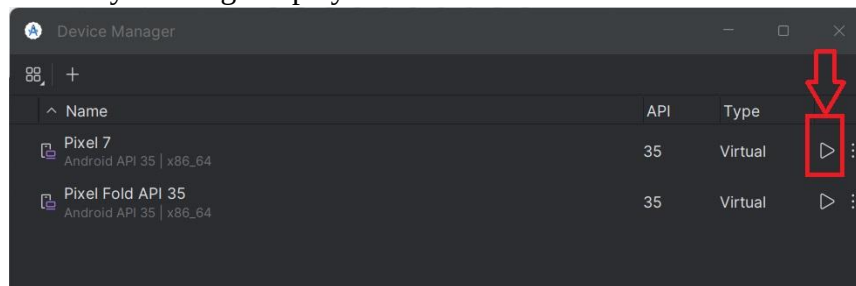
- Again, from Android Studio, expand More Actions → Click Virtual Device Manager
- Click the + sign → Proceed with Pixel 7 and click Next.



- For System Image, select API 35 and click Next



- At this point, you can rename the virtual device if you like, and set its layout style.
- Click Finish
- Run the emulator by clicking the play button.



- Now, you should be able to see an emulated phone as shown below:

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6. Let's install appium-doctor to confirm that all required steps are done correctly.

- Open your Command Prompt and type ANY of the following commands:

```
npm i appium-doctor -g
```

or

```
npm i --location=global appium-doctor
```

- Now, type `appium-doctor --version` to check appium-doctor is installed successfully. You should be able to see the version of your appium-doctor, for example, **1.16.2**.

7. We also should also install Appium Server. To do so, Open your Command Prompt and type ANY of the following commands:

```
npm i appium -g
```

or

```
npm i --location=global appium
```

Now, type `appium --version` to check appium-doctor is installed successfully. You should be able to see the version of your appium-doctor, for example, **12.11.2**.



- Restart your command prompt and then run `appium-doctor`.

You should be able to see the following options properly installed/set.

NOTE: You can ignore the following warning:

WARN AppiumDoctor ✗ android, apkanalyzer.bat could NOT be found in C:\Users\xxxxxxx\AppData\Local\Android\Sdk!

```
C:\Users>appium-doctor
[WARN] AppiumDoctor [Deprecated] Please use appium-doctor installed with "npm install @appium/doctor --location=global"
[INFO] AppiumDoctor Appium Doctor v.1.16.2
[INFO] AppiumDoctor ### Diagnostic for necessary dependencies starting ###
[INFO] AppiumDoctor ✓The Node.js binary was found at: C:\Program Files\nodejs\node.EXE
[INFO] AppiumDoctor ✓Node version is 20.15.1
[INFO] AppiumDoctor ✓ANDROID_HOME is set to: C:\Users\ \AppData\Local\Android\Sdk
[INFO] AppiumDoctor ✓JAVA_HOME is set to: C:\Program Files\Java\jdk-21
[INFO] AppiumDoctor Checking adb, android, emulator, apkanalyzer.bat
[INFO] AppiumDoctor 'adb' is in C:\Users\ \AppData\Local\Android\Sdk\platform-tools\adb.exe
[INFO] AppiumDoctor 'emulator' is in C:\Users\ \AppData\Local\Android\Sdk\emulator\emulator.exe
[WARN] AppiumDoctor ✗android, apkanalyzer.bat could NOT be found in C:\Users\ \AppData\Local\Android\Sdk!
[INFO] AppiumDoctor ✓'bin' subfolder exists under 'C:\Program Files\Java\jdk-21'
[INFO] AppiumDoctor ### Diagnostic for necessary dependencies completed, one fix needed. ###
[INFO] AppiumDoctor ### Diagnostic for optional dependencies starting ###
[WARN] AppiumDoctor ✗opencv4nodejs cannot be found.
[WARN] AppiumDoctor ✗ffmpeg cannot be found
[WARN] AppiumDoctor ✗mjpeg-consumer cannot be found.
[WARN] AppiumDoctor ✗bundletool.jar cannot be found
[WARN] AppiumDoctor ✗gst-launch-1.0.exe and/or gst-inspect-1.0.exe cannot be found
[INFO] AppiumDoctor ### Diagnostic for optional dependencies completed, 5 fixes possible. ###
[INFO] AppiumDoctor ### Manual Fixes Needed ###
[INFO] AppiumDoctor The configuration cannot be automatically fixed, please do the following first:
[WARN] AppiumDoctor →Manually install android, apkanalyzer.bat and add it to PATH. https://developer.android.com/studio
[INFO] AppiumDoctor https://developer.android.com/studio/intro/update#sdk-manager may help to setup.
```

- Now, Install UiAutomator2 Driver by Running the following command in command prompt:
`appium driver install uiautomator2`

- The final tool to be installed is Appium Inspector <https://github.com/appium/appium-inspector/releases>
 - Scroll down and select the right option for your OS.
 - Once the Inspector is downloaded, locate it and double-click on it to install it.
 - If you're faced with Window Protected Your PC window, then click on More info and choose Run anyway.
 - If not selected by default, select "Only for me (administrator)" and click Install.
 - Click Finish

NOTE: Appium's main documentation can be found here here: <https://appium.io/docs/en/2.1/>